Solution Key: Rotational Mechanics

Translation, Torque, Equilibrium, and Inertia

1 Classification of Motion

a) **Solar panel tilting on a fixed tracker:**
Pure Rotation. The object changes its orientation around a fixed axis.

b) **Heavy toolbox being hoisted vertically:**
Pure Translation. Every point on the toolbox moves the same distance in the same direction.

c) **Bicycle wheel rolling along a flat road:**
Combined Motion. The wheel rotates around its axle while the axle itself translates along the road.

2 Torque (Moment of a Force)

a) **Condition for Zero Torque:**
A force produces zero torque if the **line of action passes through the pivot point** (the distance $r = 0$ or the angle $\theta = 0°$ or $180°$).

b) **Calculation:**
Given: $F = 60\,N$, $r = 0.25\,m$, $\theta = 90°$
$M = F \cdot r \cdot \sin(\theta) = 60 \cdot 0.25 \cdot 1 = \mathbf{15\,N \cdot m}$

3 Resultant Moment (Net Torque)

Taking Counter-Clockwise (CCW) as positive:

- $M_1 = +(F_1 \cdot r) = 30 \cdot 0.2 = +6\,N \cdot m$
- $M_2 = -(F_2 \cdot r) = 45 \cdot 0.2 = -9\,N \cdot m$
- $\sum M = M_1 + M_2 = 6 - 9 = -\mathbf{3\,N \cdot m}$

Result: The net torque is $3\,N \cdot m$ in the Clockwise direction.

4 Rotational Equilibrium

For the beam to be balanced: $\sum M = 0$

- Left torque (M_1): $W_1 \cdot d_1 = 150 \cdot 1.5 = 225\,N \cdot m$ (CCW)
- Right torque (M_2): $W_2 \cdot d = 250 \cdot d$ (CW)
- Equilibrium: $225 = 250 \cdot d \implies d = \frac{225}{250} = \mathbf{0.9\,m}$

5 Moment of Inertia: Standard Shapes

For a solid cylinder: $I = \frac{1}{2}MR^2$

- $I = 0.5 \cdot 4\,kg \cdot (0.1\,m)^2 = 2 \cdot 0.01 =$ **0.02 kg · m²**

6 Moment of Inertia: Complex Systems

Total Inertia $I_{total} = I_{rod} + I_{point\_mass}$

1. $I_{rod} = \frac{1}{3}ML^2 = \frac{1}{3} \cdot 3 \cdot 2^2 = 4\,kg \cdot m^2$
2. $I_{point} = m \cdot r^2 = 1.5 \cdot 2^2 = 6\,kg \cdot m^2$
3. $I_{total} = 4 + 6 =$ **10 kg · $\mathbf{m^2}$**